

Unveiling the Sensory Symphony: Exploring the impact of Bright/Dark Haptic Stimuli on Audio and Visual Perception.

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Abstract—The present study delves into the intriguing realm of sensory perception by investigating the impact of bright and dark haptic stimuli on audio and visual perception. Haptic feedback is a sensory phenomenon that is experienced by every ordinary individual in their day-to-day lives. However, its potential influence on audio and visual senses remains unexplored to date. This research aims to unveil the intricate interplay between haptic, auditory, and visual sensations, shedding light on the cross-modal perception of stimuli characterized as bright or dark. The findings of this research are far-reaching, with potential applications in various domains, including virtual and augmented reality, multimedia design, and human-computer interaction. By uncovering the intricate relationships between haptic, audio, and visual perception, this study contributes to our understanding of the multi-sensory nature of human experience and paves the way for future advancements in sensory design and user experience

Index Terms—component, formatting, style, styling, insert

I. INTRODUCTION

Among all the expressions of timbre in the world of audio and visual perceptions one of the most researched topics happens to be brightness. Musicians, composers, and producers describe the use of brightness processing Pearce et al. (2017). Brightness is often seen as the perceptual attribute that is directly proportional to the presence of high frequencies. A tonal sample with a lot of high frequencies is perceived as brighter compared to one with less high frequencies Almeida et al. (2017). When we contemplate an auditory occurrence, we can perceive the sound waves emanating from the event through our auditory organs, as well as encounter a physical change in our skin if a considerable amount of energy is present. The propagation of sound waves can also occur through various structures, such as the surface beneath our feet or other objects we come into contact with, thereby generating a tactile response. The field of technology and science that focuses on creating and delivering tactile sensations and feedback to users is called haptics. With the advancement in technology in the fields of software, game development, and media, the

application of haptics has increased to create applications that increase user experiences by involving cross-modal sensations.

Haptics which refers to the use of tactile feedback and vibrations, has become an integral part of enhancing user experiences in various domains. Furthermore, with the advent of virtual reality (VR) and augmented reality (AR), haptic feedback has become crucial in creating a sense of presence and realism. Research has also shown that vibrotactile can be a very effective way to present music to deaf people, studies have also shown that people can detect pitch changes only with haptic signals when received by their glabrous skin surfaces. These research results also tell us about the threshold and high-frequency perception of people when hearing through the skin. Lederman & Klatzky (2004) provides a summary of the majority of the important literature. Binns (1937) used samples of wool to explore the link between visual and tactile material perception. He allowed participants to use their vision or touch to arrange six wool standards. There was no noticeable performance difference between the two modalities. Later, several research particularly examined how roughness is perceived in vision and touch. Most frequently utilized stimuli have been made of sandpaper Björkman (1967), Heller (1982), Jones & O’Neil (1985), but in BROWN (1960) and Guest & Spence (2003) cloth were also used.

II. RELATED WORK

There is evidence of research done on both visual and auditory brightness, An example of such work can be seen in the paper Cai et al. (2010). Through this paper, they presented a novel algorithm that aims to objectively evaluate the perception of "brightness" in sound. This evaluation involves analyzing various psycho-acoustic aspects such as modulation frequency, critical band, loudness level, and spectral distance. These parameters are identified as key features for the "brightness" evaluation model. The formula used for calculating the "brightness" score is a weighted combination of these parameters, reflecting their contribution to the overall

perception of brightness. To validate the algorithm and the entire design process of their experiment, the paper used both subjective and objective evaluation methods. The subjective evaluation involves listening tests where participants assess the brightness of specific musical excerpts. These subjective scores were obtained through Mean Opinion Scores (MOS) experiments. On the objective side, the algorithm is applied to the same musical excerpts. The results of the algorithm's objective evaluations are compared with the subjective scores obtained from the listening tests. The paper reported a strong correlation of 0.884 between the objective and subjective evaluations, indicating that the algorithm effectively captured the perceptual aspects of brightness that could impact the concept of auditory brightness in one's perception. However, The paper focuses solely on auditory brightness, overlooking interactions with other timbral qualities, limiting its applicability in comprehensive sound quality assessment. Expanding the algorithm's scope to encompass a broader range of sonic attributes could enhance its utility.

Another yet very interesting approach to understanding and co-relating the relationship between timbral brightness scale with frequency and spectral centroid can be seen through Schubert & Wolfe (2006). In this paper, they examined two models predicting timbral brightness based on frequency spectrum centroid. One model utilized the centroid value directly, while the other scaled it with the fundamental frequency through division. Different single-tone and pitch combinations were sequentially presented for comparison, considering intervals exceeding an octave and acknowledging pitch distinctions. Outcomes underscored a notably stronger correlation between brightness perception and frequency spectrum centroid (correlation coefficient $r = 0.513$, p less than 0.001) compared to a centroid-to-fundamental frequency ratio (correlation coefficient $r = 0.030$, $p = 0.441$). While shedding light on perceived timbral brightness, this research focuses solely on frequency-related metrics, potentially overlooking other crucial timbral characteristics. The reliance on sequential tone presentations might restrict representing real-world contexts.

After researching audio brightness, the research study went in a direction towards investigating the concept of brightness in the context of **visual** stimuli. The work of Spence (2011) guided the path to know more about the effects of timbre on visual perception through crossmodal interactions. The paper delved into whether the presence of timbre as primes would aid or hinder individuals' visual responses in tasks related to brightness and spatial texture. The investigation considered the influence of semantically congruent or incongruent timbral cues on participants' performance. Employing pre-validated natural instrument sounds and synthesized tones as stimuli in Experiment 1, the paper uncovered indications of interaction between auditory and visual senses. Incongruent audio-visual pairs led to an eight percent increase in error rates compared to congruent pairs, suggesting that the timbral quality of "brightness" influences visual perceptions of brightness, resulting in assimilation effects. Moreover, stimuli characterized as "bright" tones and brighter visual targets elicited faster

correct responses than their "dark" counterparts. However, the data suggests that the cross-modal congruency through did not significantly impact choice reaction times. While the impact on error rates is evident, the absence of significant effects on reaction times points to a more nuanced relationship between timbral cues and visual processing. Future investigations could dive deeper into the mechanisms driving these effects and consider additional factors that might contribute to the observed outcomes.

These studies all together led us to the works of Hopkins et al. (2016) where The study explores the potential of using vibrations to convey music to the skin as a means of enabling interaction among musicians, especially those with hearing impairments, during group performances. The investigation focuses on determining the thresholds for vibrotactile presentation of music on various skin areas like the fingertip, forefoot, and heel. Interestingly, no significant differences were observed between vibrotactile thresholds for different musical notes when presented to the fingertip, regardless of participants' hearing abilities. However, distinct patterns emerged in thresholds for different notes across the forefoot and heel due to spatial summation effects from the skin's receptors. Participants with normal hearing exhibited an increased awareness of high-frequency vibrations when presented slightly above the threshold, particularly noticing transient vibrations at the beginning and/or end of notes. Around 94 percent of participants reported a continuous sense of vibration within a specific note range, and based on these findings, a suitable range for vibrotactile music presentation is proposed. This range aligns with the fundamental frequencies of the human voice and various musical instruments, offering the potential for inclusive musical experiences, though managing dynamic ranges might be necessary to ensure accurate pitch perception. The results of this study led us to wonder about something fascinating: Can we sense different qualities of sound, like brightness, through touch? it raised intriguing questions like Could there potentially be timbral attributes embedded in haptic signals? The study showed that feeling musical notes on the skin is possible, but it left us curious. Is it possible to detect changes in the character of sound through touch? This inquiry challenges us to explore the potential connection between how things sound and how they feel on our skin.

III. BACKGROUND

Our efforts through this paper have been put in the direction of investigating to a very interesting yet unexplored field of study i.e., the role of haptics in creating the perception of brightness while receiving audio/visual stimuli. This paper involves two main experiments; the first experiment demonstrates the effect of bright/dark haptics on audio perception and the second experiment demonstrates the effect of bright/dark haptics on visual perception. This research was designed to see and verify the results on human perception of brightness when congruent and incongruent haptic signals were paired along with dark and bright audio or visual stimuli in a cross-modal environment.

IV. THE CROSSMODAL INTERACTION AND THE STROOP EFFECT

Crossmodal interaction, in simple terms, is when our different senses, like seeing and hearing, work together to shape how we understand the world. This happens when what we hear or see influences each other in surprising ways. For instance, people might use words usually linked to how things look or feel to describe how sounds make them feel. This mix-up of senses happens a lot when we talk about things like music and how they feel or look to us. Crossmodal correspondences might sound complex, but they're about how our senses, like seeing and hearing, work together in ways we might not expect (Marks, 1978, 2004; Spence, 2011). Imagine quickly recognizing something you see – if it's paired with a sound that doesn't seem related, you might still react faster. But if the things you see and hear connect in meaning or idea, you'll react even quicker and more accurately. For example, if a high-pitched sound comes with something shown up high, like on a screen, you'll respond faster. This happens a lot, like how certain sounds seem related to how things look, like bright sounds with bright lights (Klapetek et al., 2012; Marks, 1987; Marks et al., 1987).

Surprisingly, this hasn't been tested much when it comes to how we sense timbre. Timbre is how things sound, their unique "voice." A study by Wallmark (2019a) explored how words like "BRIGHT" or "DARK" match certain sounds. They found some connections, but it was mostly about language, not how our senses work together. Plus, we don't know the exact reasons behind these special connections when it comes to timbre. So, it's like we're exploring a new piece of the puzzle to understand how our senses mix and match in a whole new way.

The experiments in this paper are designed in a way of a Stroop effect interface where the participants were asked to respond to the different audio/ visual stimuli paired with

two different kinds of haptic signals. The Stroop effect is a fascinating cognitive phenomenon, that reveals the intricate relationship between words and colors in our perception and response. The usual setup of a Stroop interface involves presenting participants with a series of words, where the color of the word and the actual meaning it conveys may be congruent (e.g., the word "RED" displayed in red) or incongruent (e.g., the word "GREEN" displayed in red). The challenge lies in swiftly and accurately responding to the color of the presented word, irrespective of its semantic meaning. This setup creates a conflict between the automatic processing of word meaning and the directed attention required to identify the color. By analyzing participants' response times and accuracy, we gain insights into the mechanisms governing our cognitive processing and the competition between these parallel processes.

V. METHODS

A. PARTICIPANTS

- 20 university students were recruited,(9 female and 11 male).
- The mean age of these 20 students was 25.66, with the standard deviation of $SD = 1.63$. item All the participants were from different linguistic backgrounds and enrolled in various multi-disciplinary courses.
- All participants reported normal hearing and normal or corrected-to-normal vision for our experiments.

B. STIMULUI

AUDIO STIMULI: The audio stimuli used in the experiment are generated through a square wave using a band pass filter that enabled us to pass through only a specific range of frequencies, blocking both lower and higher frequencies. We established a foundation by generating two baseline frequencies, each paired with two distinct target frequencies. Anchored at a fundamental frequency of 400Hz, our baseline A was the fourth harmonic of the fundamental frequency precisely 1600HZ. The targets for the baselines were 800 Hz(second harmonic: darker target) and 2400 Hz(sixth harmonic: brighter target). The Baseline B resonated at the fifth harmonic of the fundamental(2000 Hz) and the two targets for the second baseline were at 2800 Hz(brighter target: seventh harmonic) and 1200 Hz(darker target: third harmonic).

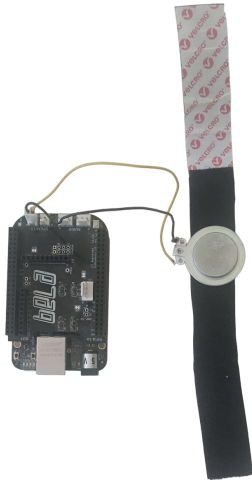


Fig. 1. HAPTIC DEVICE

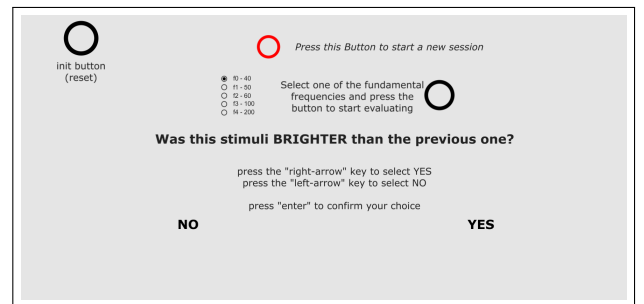


Fig. 2. Participant's View of the Pilot Interface

TABLE I

	baseline	inter stimulus interval	target + interfering stimulus	decision task	silence
Type of stimulus	Audio /Visual	silence	Audio/visual target + Haptic stimuli	silence	silence
Duration	2000ms	500ms	2000ms	Until a response is submitted (i.e. until an arrow key is pressed)	500ms

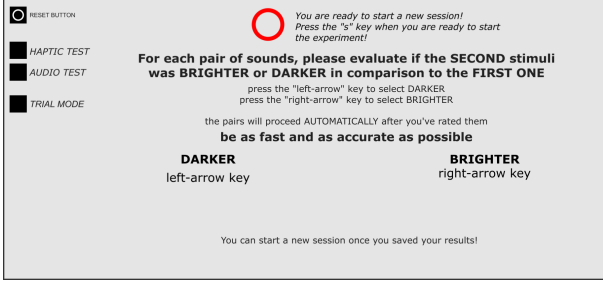


Fig. 3. Participant's View of the Audio Interface

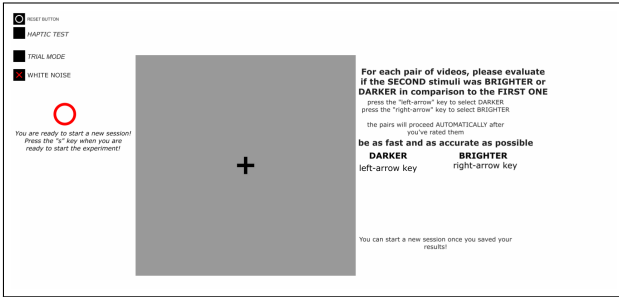


Fig. 4. Participant's View of the Visual Interface

VISUAL STIMULI: Similar to the audio stimuli, The Visual stimuli were black images of 640×640 pixels with varying opacity. The two baseline images for the visual experiment were 40 percent and 60 percent opacity, respectively. As earlier each baseline is paired with two target black squares of 10 percent more and less opacity than the baseline.

HAPTIC STIMULI: Fig 1 describes the structure of the strap used and the haptic device connected to the Bela kit. For determining two haptic stimuli a pilot experiment was conducted (described later in the paper) among a small group of people. A square wave made up of the fundamental frequency of 100Hz was finalized, the dark signal consisted of the second harmonic of the fundamental, and the bright signal was made up of the fifth harmonic of the signal.

C. APPARATUS

The experimental setup encompassed several key components. An Open Sound Control connection was established between Bela and the Max interface, facilitating seamless communication. Bela turned up as a powerful tool that offers fast and real-time handling of sound. We chose Bela for our study to make sure our audio responses were quick and smooth, allowing us to create a seamless interaction with zero

latency problems for both our audio and visual experiments. For the haptic sensory experience, a vibrotactile actuator was integrated into a band-like configuration, securely attached to the participants' wrists. This tactile interface allowed for dynamic interaction throughout the study.

The interactive platform, pivotal to the experimental process, was developed within the Max environment. Participants utilized this interface to provide their responses during the experiments. The best feature was the system's ability to systematically record participants' answers. After each trial, responses were collected and stored in the form of a structured text file, contributing to the comprehensive data acquisition process.

VI. EXPERIMENT 1: THE HAPTIC BRIGHTNESS TEST

A. PROCEDURE

During the initial study, we conducted a pilot experiment solely to determine the source of the haptic signal that would be used in the latter two experiments, the main aim of this experiment was to determine two haptic signals that can be distinctly understandable by the participants as a dark and bright stimulus, the design of the test involved 5 fundamental frequency(40,50,60,100 and 200 Hz) and for each frequency, there were nine different centroid values made up of their respective harmonics(Fig 2 shows the design of the interface made for this experiment). For every fundamental signal, a step-wise comparative analysis was done by the participants where they were asked to compare the current haptic stimulus with the previous, two buttons were initialized the right arrow button was meant to press when there is a difference in the stimuli i.e. they feel that the recent stimuli were brighter than the previous one and they were meant to stop at the point by pressing the left arrow key when they couldn't hear any difference between the two consecutive haptic stimuli i.e. both stimuli feel like they are at the same brightness level.

B. RESULTS

This pilot experiment was performed within a small group of 5 people, after carefully analyzing the report we could see that the stimuli made up of the 100 Hz fundamental frequency had quite a significant impact on the participants, the results consistently said that the participants could distinguish up to the fifth harmonic of the signal. Hence as the source of the dark signal, we chose to go with the second harmonic of the fundamental, and for the bright haptic signal we finalised the fifth harmonic of the fundamental signal.

VII. EXPERIMENT 2: THE AUDIO HAPTIC INTERFERENCE

A. PROCEDURE

The experiment was conducted in an acoustical-treated quiet room on a computer with a screen size of 14 inches with 1366x768 pixel resolution. The audio stimuli were delivered to participants through Audio Technica ATH-M50X headphones, which were connected to the computer using a Scarlet Solo audio interface. (Fig3 gives us an insight into how the interface looked) Before the start of each experiment, after a short greeting to the participants, A brief description of the entire experimental procedure was enacted to them. They were asked to use their dominant arm to give the response for the trials and we would attach our haptic actuator on the wrist of the nondominant arm. Sincere care was taken to maintain the absolute comfort of the participants in terms of the sitting position and other physical modalities of the test environment. The designed interactive interface gave us the chance to give the participants to experience random audio and haptic stimuli and to get used to the experiment, the audio level of the headphones was adjusted according to the participant's comfort level. In addition to that they were given 6 practice trials that exactly replicated the actual test procedure to familiarize themselves with the actual task. Participants during the actual experiment at first heard the baseline audio tone for 2000ms followed by a silence of 500ms, after this, they heard the target tone paired with a haptic stimulus for 2000ms. The procedure is presented in the above table 1. The target audio tone heard by the participants was either brighter, darker, or identical to the baseline audio tone.

- The two baseline audio stimuli had two targets(one brighter and one darker).
- 2) Along with that we also had a control condition for each baseline where the baseline repeats itself as the target.
- 3) And for every possible combination of the baseline and the target we had two interfering haptic stimuli.
- 4) which gives us a total of 12 possible combinations.
- 5) Because each condition was repeated 5 times it gave us a total of 60 trials altogether.

The order of the entire trials was randomized hence the participants were informed that every trial is different from the previous one and hence to notice the subtle changes in the targets and the base lines they should hear the audio tones carefully. They were asked to answer as quickly and accurately as they could by pressing on any one of the arrow keys of the computer keyboard (left arrow key for Darker and right arrow key for brighter). The experiment took approximately 15 minutes.

B. RESULT

The analysis was conducted using Python 3.6 available with Google Colab. For access to the various data collected and Python scripts, please refer to the Supplementary Materials provided. Adhering to the previous studies, outlier thresholds

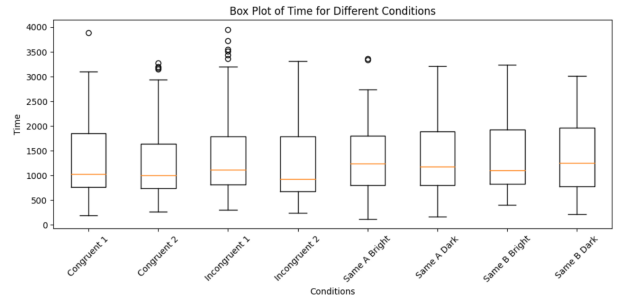


Fig. 5.

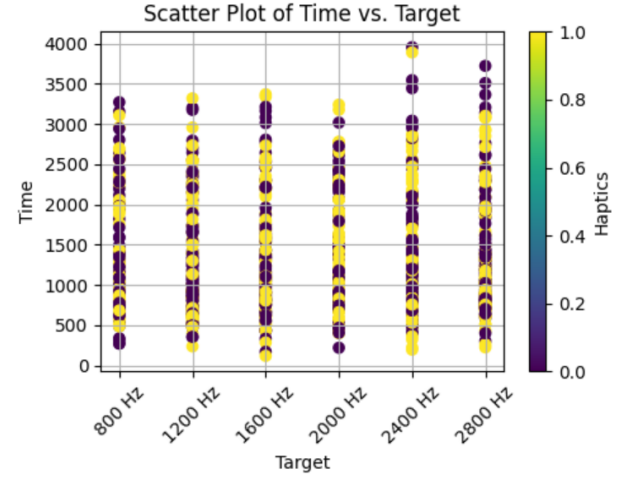


Fig. 6.

were implemented on the reaction time (RT) data, resulting in the exclusion of responses below 100 ms or above 4000 ms. In addition to that, a second layer of evaluation was performed identifying participants with error rates surpassing chance levels (more than 50 percent), leading to the exclusion of their data. This action led to the removal of 382 responses, constituting 25 percent percent of the total dataset comprising 1200 responses. A comprehensive summary of the obtained results is presented in Table 2. This summary constitutes metrics such as mean reaction times for different Congruent and In-congruent interference conditions standard deviation and error percentage aggregated across all the participants.

Following the primary analysis the experiment demanded two types of analysis, 1) reaction time for different congruencies and 2) accuracy of the participants in different scenarios. We divided the entire data set into two types, the first one consisting of all the congruent and incongruent interference scenarios where the baseline and the target are not the same, and the second set of data consisted of all the data sets where the target and the baseline are the same. Because of the nature of our data where repeated analysis was performed and also to satisfy our interest in analyzing the fixed effects and random effects (variations specific to individual subjects or groups), mixed linear models allowed us to account for the hierarchical structure of the data and provide more accurate estimates of the

TABLE II

Target	HAPTIC STIMULI	Congruency	Mean Reaction Time (milliseconds)	Standard Deviation	Error Percentage
Brighter Audio	Bright Haptic	Congruent	1322.69	754.59	2.469
Darker Audio	Dark Haptic	Congruent	1241.96	701.544	2.020
Brighter Audio	Dark haptic	Incongruent	1367.99	782.654	4.264
Darker Audio	Bright Haptic	Incongruent	1243.14	719.874	3.815
Same condition A	Bright Haptic	-	1371.03	742.094	2.469
Same condition B	Bright Haptic	-	1395.85	736.835	2.244
Same condition A	Dark Haptic	-	1408.94	795.598	2.581
Same condition B	Dark Haptic	-	1369	727.403	3.591

effects while addressing potential sources of variability. (Fig 6 a scatter plot, gives us an insight into the time responses against different targets and the haptic signal and Fig 5 a box plot gives us an insight into the time responses against the different combined combinations between the target, base, and the haptic stimuli)

Analysis of the reaction time of the first subset(excluding the control condition): As a dependent variable we used the logarithm of the reaction time and analyzed it against the target and the haptic signal, coefficients of predictor variables showed slight variations, such as Intercept: Coefficient = 7.026, coefficient of the darker target = -0.058, coefficient of the dark haptic = 0.055, along with the interaction between the two predictor variables target[T.Darker]: haptic[T.Dark]: Coefficient = -0.049. which signified the effect of these values on the reaction time, but as the values were not significant we couldn't see any effect of these conditions on the reaction time. In the accuracy analysis, because we were dealing with a binomial set of data our study suggested the use of Generalized linear mixed model with the use of Bayesian methods, we set the accuracy as the dependent variable against the predictor variables target*haptic that tells us about the effect of the interaction between both target and the haptic stimuli. In addition to that, we also included the Participant ID as the random effect term. The result of the **accuracy model** showed some significance of different parameters that could have potentially affected the accuracy of the participants. The coefficient value of the target darker was -0.600 with a P-value of 0.058, for the Dark haptic signal the coefficient value was -0.782 with a p-value of 0.012, and for the interaction between the darker target and dark haptic stimuli, the coefficient value was 1.630 with the P-value less than 0.0001. It can be seen there were marginal effects of both the target and haptic dark and a significant effect of the interaction between the darker target and the dark haptic stimuli. **Accuracy analysis of the second subset(including only the control condition:** In the audio haptic interference experiment, there was another interesting class of data i.e. the scenarios when both the baseline stimuli and the target stimuli were the same, We tested the accuracy model for it, this time the dependent variable was the accuracy of the participants against the haptic variable as the predictor and the participant ID as the random effects parameter. We

were successful in noticing marginal effects by the haptic dark stimuli but not any other significant parameter that could affect the accuracy outcome of the participants.

VIII. EXPERIMENT 3: THE VISUAL HAPTIC INTERFERENCE

A. PROCEDURE

The visual haptic interference experiment was also conducted in the same space as the former experiment. The same set of participants were also asked to perform the visual haptic interference as well. We divided the total number of participants into 2 equal groups(10 each) the first group of participants took part first in the audio haptic interference followed by the Visual Haptic Interference experiment where as the other half performed first the Visual Haptic Interference experiment followed by the Audio Haptic Interference experiment. This was done to potentially eliminate any sort of bias that could have occurred due to the sequence of the experiments. But this time in place of audio tones they were played a white noise through the Audio Technica ATH-M50X headphones, which were connected to the computer using a Scarlet Solo audio interface(Fig 3 shows us the interactive interface that was used for the experiment) . The audio level of the white noise in the headphones was adjusted according to the participant's comfort. This was done in order to avoid any sound that might emit from the vibrotactile actuator during the test. They were also given around 6 practice trials that replicated the actual test procedure to familiarize themselves. Participants during the Visual Haptic Interference experiment first saw the baseline visual square (either with an opacity of 40 percent or 60 percent) for 2000ms followed by a silence of 500ms, after this they saw the target visual square paired along with a haptic stimulus for 2000ms. The procedure is presented in Table 1 for further reference. The target visual stimuli displayed to the participants were either brighter, darker, or identical to the baseline visual square opacity. Similar to the former experiment the Visual Interference Haptic experiment also had 60 trials for each participant.

- There were two baseline Visual stimuli that had two targets(one brighter and one darker).
- The control condition for each baseline visual stimulus repeats itself as the target.

TABLE III

Target	HAPTIC STIMULI	Congruency	Mean Reaction Time (milliseconds)	Standard Deviation	Error Percentage
Brighter Opacity	Bright Haptic	Congruent	1377.88	667.318	7.1949
Darker Opacity	Dark Haptic	Congruent	1278.83	713.469	5.555
Darker Opacity	Brighthaptic	Incongruent	1364.03	697.242	13.1148
Brighter Opacity	Dark Haptic	Incongruent	1327.21	737.791	11.3843
Same Opacity A	Bright Haptic	-	1433.47	683.189	4.189
Same Opacity B	Bright Haptic	-	1376.15	779.051	11.02
Same Opacity A	Dark Haptic	-	1462.51	682.287	6.55
Same Opacity B	Dark Haptic	-	1203.46	691.618	0.364

- For every possible combination of the baseline and the target stimuli we had two interfering haptic stimuli.
- Which again gave us a total of 12 possible combinations.
- Each condition was repeated 5 times hence it gave us a total of 60 trials.

B. RESULT

The analysis of the Visual Haptic stimuli also followed the same outlier filtering method reaction time (RT) data, the responses below 100 ms or above 4000 ms were excluded, and the second layer of the evaluation was performed identifying participants with error rates surpassing chance levels (less than 50percent), leading to the exclusion of their data. This action led to the removal of responses, constituting 8.5 percent of the total data set. A comprehensive summary of the obtained results is presented in Table 3. The data was divided into two subsets the first subset had all the congruent and incongruent conditions except the trials where the baseline opacity was identical to the target stimulus opacity. The first subset was analyzed both in terms of reaction time and accuracy whereas the second subset was analyzed only through the accuracy model. (Fig 7 a scatter plot gives us an insight into the time responses against different targets and the haptic signal and Fig 8 a box plot gives us an insight into the time responses against the different combined combinations between the target, base, and the haptic stimuli)

The first model measured the reaction time of the participants, A Linear Mixed effects model was used to analyze the change in the reaction times of the participants, the dependent variable set for this model was the logarithm of the reaction time, and the predictor variable were the interference effect of the target and the haptic stimuli. **Analysis of the first subset** showed us that the coefficient value for the Darker Target was at -0.045 and the dark haptic stimuli are -0.045 but they are not statistically significant as the P values are at 0.320 and 0.317 respectively, indicating that the changes in these variables do not have a significant impact on the log of time. Similarly, the interaction term "target[T.Darker]: haptic[T.Dark]" also turned out to be statistically insignificant, suggesting that the combined effect of the "target" and "haptic" conditions does not have a significant influence on the log of time. While in the first subset, the **accuracy analysis** was performed using the

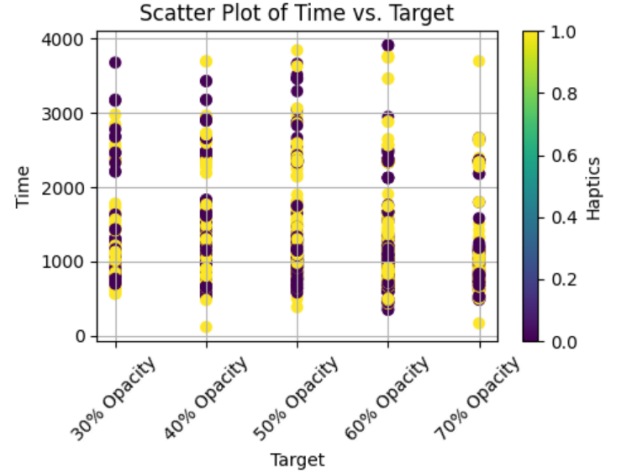


Fig. 7.

Bayesian approach of the generalized linear mixed model, the coefficient value of the predictor variables had some variations, the darker visual target coefficient was at 0.730, the dark haptic stimuli had a coefficient of -0.131 and the interaction component between the darker stimuli and the dark haptic signal had a coefficient of -0.147, looking at the significance of all these parameters we can say that there was a marginal effect of the dark visual stimuli signal as the P-value of the term was at 0.031.

Accuracy Analysis of the second subset(includes all the conditions where target = baseline): In this type of data set we only wanted to see the effect of haptics on the accuracy as both the target and the baseline are the same, in the formula of the model the accuracy term was set as the dependent variable against the haptic stimuli as the predictor variable with the participant Id to identify any random effect that might occur from within the participants. It was clearly seen that the dark haptic stimuli had a significant effect on the accuracy with a coefficient of 1.366 and a P-value less than 0.001

IX. DISCUSSION

The present study aimed to investigate the crossmodal interactions between different sensory modalities, specifically focusing on the effects of haptic and brightness, audio and

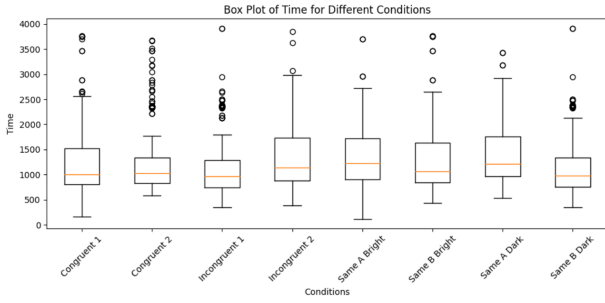


Fig. 8.

haptic interference, as well as vision and haptic interference, on participants' perceptual experiences. The findings provide valuable insights into how sensory modalities can influence and modulate one another, shedding light on the complex nature of multi-sensory perception.

Haptic and brightness The haptic and brightness experiment done initially as a part of a pilot experiment helped us in designing the entire experiment, the participant's feedback in refining haptic stimuli was a very crucial contribution towards the selection of optimal fundamental frequency and the harmonics to ensure the clarity and effectiveness of haptic signals, thereby facilitating meaningful crossmodal interactions. This pilot-driven approach contributes to the overall validity and relevance of the experimental setup. We conducted the experiment with five different fundamental frequencies, and for each of these frequencies, we compared nine different centroid frequencies. The outcomes of these comparisons led us to the conclusion that most of the participants were capable of distinguishing the up to the fifth harmonic of the 100Hz fundamental frequency and this led us to using them as the bright and the dark interfering signals. **Audio and Haptic Interactions** The results of the audio and haptic experiment revealed intriguing patterns of interaction between auditory and tactile stimuli. The observed correlation between the modulation frequency and participants' subjective evaluation of "brightness" points toward a potential link between auditory modulation and tactile perception. This suggests criteria where the crossmodal integration mechanism between audio timbre perception and tactile sensations can interfere with each other. We couldn't see any crossmodal effect or random effect in terms of reaction times like Wallmark et al. (2021), this might have occurred due to the nature of the experiment where the participants were supposed to answer as quickly and as accurately as they could because of the time factor the potential effects were unseen in our case. But at the same time, certain features also turned out to be significant and came to light despite the data set being very small. The careful analysis of the result summary of the accuracy model suggested that the interference effect between the darker audio stimuli and the dark haptic stimuli was quite effective in increasing the accuracy percentage of the participants. This mainly happened in one of the congruent scenarios where the target became darker than the baseline and along with that the haptic stimuli

given to them were also dark, which in turn shows us that the participants could understand the change in audio brightness better and effectively with the addition of the haptic stimulus.

Video and Haptic Interference In the context of the vision and haptic experiment, as we progressed with the investigation we could see that the variation of errors and the reaction time showed a little variation from the former tasks. The accuracy analysis showed us the interaction component between darker stimuli and dark haptic signal didn't impact the accuracy much, but on the other hand, the darker target stimuli indicated some marginal effect on the accuracy of the participants. Analyzing the second subset of this experiment, which included conditions where the target equaled the baseline, the focus shifted to exploring the impact of haptics on accuracy. In this model, accuracy served as the dependent variable against haptic stimuli as the predictor variable. Incorporating participant IDs helped identify any potential random effects within participants. The analysis revealed an impact of dark haptic stimuli on the accuracy, in both the subsets of the Vision experiment

Modeling and Analysis The utilization of Generalized Linear Mixed Models (GLMMs) for analyzing the collected data offered a robust framework for exploring the relationships between predictor variables and the dependent variables. The incorporation of random effects accounts for individual variability, yielding a more accurate representation of real-world complexities. Furthermore, the Bayesian methods employed in the GLMM analysis provided a powerful tool for estimating model parameters and uncertainty, enhancing the reliability of our findings.

X. FUTURE WORKS

The outcomes of our experiments, focused on haptic and crossmodal interactions, open the door to several promising options for future research. Building upon the success of the pilot-driven haptic experiment, further investigations can explore ways to refine the haptic stimuli even more effectively. Delving deeper into the perceptual nuances of fundamental frequencies and harmonics could lead to the identification of optimal configurations that maximize clarity and meaningfulness in crossmodal interactions. Additionally, incorporating advanced techniques, such as psychophysical scaling, could provide deeper insights into the perceptual boundaries and sensitivities of participants when combining auditory and tactile sensations.

While our audio and haptic experiment revealed intriguing trends in the relationship between auditory and tactile stimuli, future research can dive deeper into the complexities of crossmodal integration mechanisms. Further investigation can be done into the reasons behind the absence of significant effects in reaction times, despite some notable trends in terms of accuracy, which could shed light on the relation between cognitive processes and temporal limitations. This could involve examining scenarios where participants are allowed more time to respond, enabling the exploration of subtle crossmodal effects. The feedback of the participants from

the visual and haptic interference experiment highlights the potential challenges participants faced in comparing brightness across sensory modalities. Future investigations could aim to pinpoint the sources of these difficulties, potentially involving participants with different visual tasks or providing additional training to minimize the errors.

Expanding beyond individual modalities, the research could look into the intricate fusion processes that govern multisensory perception. Integrating the findings from our experiments into a broader and new framework that considers audio, haptic, and visual modalities combined together could yield deeper insights into the principles governing how the human brain combines sensory inputs to form a coherent perceptual experience. Our study provided some valuable groundwork for future investigations that aim to unravel the intricate nature of crossmodal interactions. The interdisciplinary nature of our work opens doors for collaborations across fields like psychology, cognitive science, and engineering, enabling a comprehensive exploration of multisensory perception and interaction. Future research could also look into the neural underpinnings of these crossmodal interactions using advanced neuroimaging techniques. Investigating the role of individual differences in sensory perception, such as age, gender, and sensory acuity, could provide a more comprehensive understanding of multisensory integration.

XI. CONCLUSION

In conclusion, our study provides valuable insights into the intricate interplay between sensory modalities, highlighting the interconnectedness of auditory, visual, and haptic cues. The observed effects of crossmodal interactions offer a glimpse into the dynamic nature of human perception, which may pave the way for innovative applications in various fields. As we continue to unravel the complexities of multisensory perception, further research holds the promise of enriching our understanding of how our senses collaborate to shape our perceptual reality.

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